

CEG 4750/6750, Information Security Homework 3

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Question 1. (5 points) Find integer x such that $3 \cdot x \equiv 1 \pmod{23}$.

Answer: $x = 8$ because $3 \cdot 8 = 24$ and $24 \bmod 23 = 1$

Question 2 (5 points). Using Fermat's theorem to compute $3^{139} \bmod 11$.

Answer: So, if p is prime and a is not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$.

Here, $p = 11$, so $3^{10} \equiv 1 \pmod{11}$.

We need to compute $3^{139} \bmod 11$:

$$139 \div 10 = 13 \text{ remainder } 9$$

$$\text{So } 3^{139} = 3^{10 \times 13 + 9} = (3^{10})^{13} \times 3^9 \equiv 1^{13} \times 3^9 \pmod{11}$$

Now compute $3^9 \bmod 11$:

$$3^2 = 9 \equiv 9 \pmod{11}$$

$$3^4 = (3^2)^2 = 9^2 = 81 \equiv 4 \pmod{11} \text{ [since } 81 - 77 = 4]$$

$$3^8 = (3^4)^2 = 4^2 = 16 \equiv 5 \pmod{11}$$

$$3^9 = 3^8 \times 3 \equiv 5 \times 3 = 15 \equiv 4 \pmod{11}$$

Therefore, $3^{139} \bmod 11 = 4$.

Question 3 (5 points) Compute the Euler Totient function of 200, i.e., $\phi(200)$.

Answer: first, factor 200: $200 = 2^3 \times 5^2$

For $\phi(n)$ where $n = p^a \times q^b$:

$$\phi(n) = n \times (1 - 1/p) \times (1 - 1/q)$$

$$\phi(200) = 200 \times (1 - 1/2) \times (1 - 1/5)$$

$$\phi(200) = 200 \times (1/2) \times (4/5)$$

$$\phi(200) = 200 \times (4/10) = 200 \times 0.4 = 80$$

Alternative formula: $\phi(p^a \times q^b) = p^{a-1}(p-1) \times q^{b-1}(q-1)$

$$\phi(200) = 2^{3-1}(2-1) \times 5^{2-1}(5-1)$$

$$\phi(200) = 2^2 \times 1 \times 5^1 \times 4$$

$$\phi(200) = 4 \times 5 \times 4 = 80$$

Thus, the answer $\Rightarrow \phi(200) = 80$

Question 4 (5 points) Compute the Euler Totient function of 720, i.e., $\phi(720)$.

Answer: first, factor 720:

$$720 = 72 \times 10 = (8 \times 9) \times (2 \times 5) = 2^3 \times 3^2 \times 2 \times 5$$

$$\begin{aligned}
720 &= 2^4 \times 3^2 \times 5^1 \\
\phi(720) &= 720 \times (1 - 1/2) \times (1 - 1/3) \times (1 - 1/5) \\
\phi(720) &= 720 \times (1/2) \times (2/3) \times (4/5) \\
\phi(720) &= 720 \times (1/2 \times 2/3 \times 4/5) \\
\phi(720) &= 720 \times (8/30) = 720 \times (4/15) \\
\phi(720) &= 720 \times 4 \div 15 = 2880 \div 15 = 192
\end{aligned}$$

Using the alternative formula:

$$\begin{aligned}
\phi(720) &= 2^{(4-1)}(2-1) \times 3^{(2-1)}(3-1) \times 5^{(1-1)}(5-1) \\
\phi(720) &= 2^3 \times 1 \times 3^1 \times 2 \times 5^0 \times 4 \\
\phi(720) &= 8 \times 3 \times 2 \times 4 = 8 \times 24 = 192
\end{aligned}$$

Thus, our answer $\Rightarrow \phi(720) = 192$

Question 5 (10 points) Compute $\phi(45)$; Compute $7^{48} \bmod 45$ using Euler's Theorem.

Answer: For $\phi(45)$

$$\begin{aligned}
\text{Factor } 45 &\Rightarrow 45 = 9 \times 5 = 3^2 \times 5^1 \\
\phi(45) &= 45 \times (1 - 1/3) \times (1 - 1/5) \\
\phi(45) &= 45 \times (2/3) \times (4/5) \\
\phi(45) &= 45 \times (8/15) = 45 \times 8 \div 15 = 360 \div 15 = 24
\end{aligned}$$

Thus, the answer being $\phi(45) = 24$

Now, for $7^{48} \bmod 45$, Euler's Theorem implies that if $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$

Here, $\gcd(7, 45) = 1$ and $\phi(45) = 24$, so $7^{24} \equiv 1 \pmod{45}$

Now, $7^{48} = 7^{(24 \times 2)} = (7^{24})^2 \equiv 1^2 \equiv 1 \pmod{45}$

Hence, the answer $\Rightarrow 7^{48} \bmod 45 = 1$

Question 6 (5 points) Let $Z_n^* = \{x \mid 1 \leq x \leq n-1, \text{ where } \gcd(x, n)=1\}$. $Z_{20}^* = ?$

Answer: first, we find all numbers from 1 to 19 that are coprime to 20 (i.e., share no common factors with 20 other than 1).

$$20 = 2^2 \times 5. \text{ So numbers must not be divisible by 2 or 5.}$$

Numbers 1-19:

- Divisible by 2: 2, 4, 6, 8, 10, 12, 14, 16, 18 (9 numbers)
- Divisible by 5: 5, 10, 15 (10 already counted)

Numbers coprime to 20: 1, 3, 7, 9, 11, 13, 17, 19

Hence, the answer: $*Z_{20} = \{1, 3, 7, 9, 11, 13, 17, 19\}^{**}$